

Comparison to Existing Tools I

Comparison to Existing Tools II

The gap analysis established that no single tool integrates all six cross-cutting concerns. Here we synthesize the fourteen-dimension comparison into three structural insights. First, the landscape bifurcates: workflow managers (Snakemake [koster2012snakemake], Nextflow [ditommaso2017nextflow], CWL [amstutz2016cwl]) provide distributed execution but no manuscript support; publication tools (Quarto [allaire2024quarto], Jupyter Book, R Markdown [xie2018dynamic], Typst [madje2023typst], Overleaf [overleaf2025], Prism [openai2026prism]) author documents but embed no integrity guarantees; and DVC [iterative2024dvc] versions artifacts without orchestrating pipelines. `template/` occupies the intersection, sacrificing distributed execution for unified enforcement of testing, provenance, and documentation. This positioning is complementary—a mature deployment might use Nextflow upstream and `template/` for rendering, testing enforcement, and provenance downstream. Typst, with its faster compilation cycle, could serve as an alternative rendering backend if a Pandoc writer were contributed.